

① Medical Image Enhancement

1. It would recommend Gamma

(power-law) Transformation with $\gamma < 1$.

2. Gamma transformation increases the brightness of dark region without greatly affecting already bright region in an x-ray image, bones already bright, but soft tissues appear dark. This transformation makes the tissues more visible helping doctors detect abnormalities easily.

3. The darker tissues become to lighter and clear. The overall image quality improves, making both bones and soft tissues easier to examine.

2. Night - Time Surveillance

1. Log Transformation is the best choice.

2. Since most pixel values are very low (0-40) the image is too dark. Log transformation expands these low intensity values making the image brighter. Negative transformation only reverse the brightness values and doesn't improve the visibility of dark area.

3. The enhanced image will be brighter with better visibility. Objects such as car, people, roads and parking lines will become clearer making surveillance more effective.

3. Finger print Recognition:

1. constraint stretching:

2. Finger print images often have low constraint between ridges and valleys. constraint stretching increases the difference b/w these gray levels making the fingerprint pattern shapes and easier to recognise.

3. Log Transformation would brighten the dark parts to the fingerprint but it would not improve the ridge and valley contrast as effectively as constraint stretching. As a result, some fingerprint details may become less distinct reducing recognition accuracy.